

The Syllabus for Basic Academic Abilities in the EJU
(To be applied to the questions of the 2026 EJU 1st Session)

〈 **Mathematics Syllabus** 〉

(Correspondence with the Course of Study for high schools is attached)

[Purpose of the Examination]

The purpose of this examination is to test whether international students have the basic academic ability in mathematics necessary for studying at universities or other higher educational institutions in Japan.

[Classification of the Examination]

There are two courses. Course 1 is for undergraduate faculties and departments for which a basic knowledge of mathematics is considered necessary. Course 2 is for undergraduate faculties and departments for which a knowledge of more advanced mathematics is necessary.

At the time of taking the examination, the examinee must choose whether to take Course 1 or Course 2; the examinees should follow the instructions given by the university or the department to which they are applying.

[Symbols and Terminology]

The symbols are those used in Japanese high school textbooks; the English version of the test uses standard English terms, and the Japanese version of the test uses the terms used in Japanese high school textbooks.

[Scope of Questions]

The topics covered by the examination are shown below.

- **The Course 1 examination covers only topics 1 to 7**
- **The Course 2 examination covers all topics, 1 to 20**

The topics are covered by the standard text books (Mathematics I, Mathematics II, Mathematics III, Mathematics A, Mathematics B, Mathematics C) used in Japanese high schools.

What is taught in Japanese elementary and junior high schools is regarded to have been already learned and is therefore included in the scope of this examination.

〈 **Topics** 〉

1. Numbers and expressions: Mathematics I

(1) Calculation of expressions

- 1) Addition, subtraction and multiplication of polynomials
- 2) Factorization

(2) Real numbers

- 1) Real numbers
- 2) Calculation of expressions containing square root

(3) Linear inequalities

- 1) Equations and inequalities containing absolute values

2. Sets and propositions: Mathematics I

- (1) Sets
- (2) Propositions and conditions

3. Quadratic functions: Mathematics I

- (1) Quadratic functions and their graphs
 - 1) Graphs of quadratic functions
 - 2) Maximum and minimum values of quadratic functions
 - 3) Determination of quadratic functions
- (2) Quadratic equations and inequalities
 - 1) Solutions of quadratic equations
 - 2) Graphs of quadratic functions and quadratic equations
 - 3) Graphs of quadratic functions and quadratic inequalities

4. Figures and measurements: Mathematics I

- (1) Trigonometric ratios
 - 1) Sine, cosine, tangent
 - 2) Relations between trigonometric ratios
- (2) Trigonometric ratios and figures
 - 1) Law of sine, law of cosine
 - 2) Measurement of figures (including application to solid figures)

5. Number of possible outcomes and probability: Mathematics A

- (1) Number of possible outcomes
 - 1) Principles of counting (including the number of elements of a set, addition principle, multiplication principle)
 - 2) Permutations, combinations
- (2) Probability
 - 1) Probability and its fundamental properties
 - 2) Independent trials and probability
 - 3) Binomial probability
 - 4) Conditional probability
 - 5) Expected value

6. Properties of integers: Mathematics A

- (1) Divisors and multiples
 - 1) Determination of multiples
 - 2) Prime numbers and prime factorization
 - 3) Greatest common divisor, least common multiple
 - 4) Division of integers

- (2) Euclidean algorithm
- (3) Linear Diophantine equations
- (4) Base n notation

7. Properties of figures: Mathematics A

- (1) Plane figures
 - 1) Properties of triangles
 - Ratios of sides of a triangle
 - Five centers of a triangle (circumcenter, incenter, centroid, orthocenter and excenter)
 - Ceva's theorem, Menelaus' theorem
 - 2) Properties of circles
 - Quadrilaterals inscribed in a circle
 - Circles and lines
 - Power of a point theorem
 - Relative position of two circles
- (2) Figures in space
 - 1) Lines and planes
 - 2) Polyhedrons

8. Various Expressions: Mathematics II

- (1) Expressions and proofs
 - 1) Division of polynomials, fractional expressions, binomial theorem, identities
 - 2) Proofs of equalities and inequalities
- (2) Higher-degree equations
 - 1) Complex numbers and solutions of quadratic equations
 - 2) Relation between roots and coefficients
 - 3) Remainder theorem and factor theorem
 - 4) Solutions of higher-degree equations and properties

9. Figures and equations: Mathematics II

- (1) Points and lines
 - 1) Coordinates of a point
 - 2) Equations of straight lines
 - 3) Distance between a point and a line
- (2) Circles
 - 1) Equation of a circle
 - 2) Relation between a circle and a line
- (3) Locus and region
 - 1) Locus and equations
 - 2) Region defined by inequalities

10. Exponential and logarithmic functions: Mathematics II

- (1) Exponential functions
 - 1) Extension of exponents
 - 2) Exponential functions and their graphs
- (2) Logarithmic functions
 - 1) Properties of logarithms
 - 2) Logarithmic functions and their graphs
 - 3) Common logarithms

11. Trigonometric functions: Mathematics II

- (1) Trigonometric functions
 - 1) General angle and radian
 - 2) Trigonometric functions and their basic properties
 - 3) Trigonometric functions and their graphs
- (2) Addition formulas
 - 1) Trigonometric functions and addition theorems
 - 2) Applications of the addition theorem
 - 3) Transformation of trigonometric functions

12. Sequences of numbers: Mathematics B

- (1) Sequences and their sums
 - 1) Arithmetic progressions and geometric progressions
 - 2) Differences sequence
 - 3) Various sequences
- (2) Recurrence relations and mathematical induction
 - 1) Recurrence relations and sequences
 - 2) Mathematical induction

13. Probability distributions: Mathematics I, Mathematics B

- (1) Random variables and probability distributions
- (2) Expected value, variance, standard deviation
- (3) Covariance, correlation coefficient
- (4) Transformations of random variables
- (5) Sum of random variables and its expected value

14. Vectors: Mathematics C

- (1) Vectors on a plane
 - 1) Vectors and their operations
 - 2) Components of vectors

- 3) Inner products of vectors
 - (2) Vectors and plane figures
 - 1) Position vectors
 - 2) Vectors and figures
 - 3) Vector equations
 - (3) Space coordinates and vectors
 - 1) Space coordinates
 - 2) Spatial vectors and their operations, inner product
 - 3) Components of vectors
 - 4) Vectors and spatial figures
 - 5) Vector equations
15. Complex plane: Mathematics C
- (1) Complex plane
 - 1) Complex plane
 - 2) Polar form of complex numbers
 - (2) De Moivre's theorem
 - (3) Complex numbers and figures
16. Equations and curves: Mathematics C
- (1) Conic curves (parabolas, ellipses, hyperbolas)
 - (2) Parameterization
 - (3) Polar coordinates and polar equations
17. Functions: Mathematics III
- (1) Rational functions and irrational functions
 - (2) Inverse functions and composite functions
18. Limits: Mathematics III
- (1) Sequences and their limits
 - 1) Limits of sequences
 - 2) Sums of infinite series
 - (2) Functions and their limits
 - 1) Limits of functions
 - 2) Continuity of functions
19. Differential calculus: Mathematics II, Mathematics III
- (1) Derivatives
 - 1) Differential coefficients and derivatives
 - 2) Derivatives of the sum/difference/product/quotient of functions

- 3) Derivatives of composite functions, derivatives of inverse functions
- 4) Derivatives of trigonometric, exponential, and logarithmic functions
- 5) nth derivative

(2) Applications of the derivative

- 1) Tangents
- 2) Increase and decrease in function value
- 3) Maximum and minimum values of a function
- 4) Graphs of a functions
- 5) Velocity, acceleration
- 6) Approximate expressions

20. Integral calculus: Mathematics II, Mathematics III

(1) Indefinite and definite integrals

- 1) Integrals and their basic properties
- 2) Integration by substitution, integration by parts
- 3) Integrals of various functions

(2) Applications of the integral

- 1) Area
- 2) Volume
- 3) Length of a curve